Chapter 8

Trigonometric Functions

Only One Option Correct Type Questions

1. The number of solutions of the pair of equations
$2\sin^2\theta - \cos 2\theta = 0$
$2\cos^2\theta - 3\sin\theta = 0$
in the interval $[0, 2\pi]$ is **[IIT-JEE-2007 (Paper-1)]**
(A) Zero (B) One
(C) Two (D) Four

2. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A$ is **[IIT-JEE-2010 (Paper-1)]**
(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$
(C) 1 (D) $\sqrt{3}$

3. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals
[IIT-JEE-2012 (Paper-2)]
(A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$
(C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

4. In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the inradius to the circum-radius of the triangle is
[JEE (Adv)-2014 (Paper-2)]
(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$
(C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

5. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has
[JEE (Adv)-2014 (Paper-2)]
(A) Infinitely many solutions (B) Three solutions
(C) One solution (D) No solution

6. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

[JEE (Adv)-2016 (Paper-1)]

(A) $2(\sec\theta - \tan\theta)$
(B) $2\sec\theta$
(C) $-2\tan\theta$
(D) 0

7. Let $S = \left\{x \in (-\pi, \pi) : x \neq 0, \pm\frac{\pi}{2}\right\}$. The sum of all distinct solutions of the equation $\sqrt{3}\sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to **[JEE (Adv)-2016 (Paper-1)]**

(A) $-\frac{7\pi}{9}$
(B) $-\frac{2\pi}{9}$
(C) 0
(D) $\frac{5\pi}{9}$

8. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to **[JEE (Adv)-2016 (Paper-2)]**

(A) $3 - \sqrt{3}$
(B) $2(3 - \sqrt{3})$
(C) $2(\sqrt{3} - 1)$
(D) $2(2 + \sqrt{3})$

One or More Option(s) Correct Type Questions

9. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then **[IIT-JEE-2008 (Paper-1)]**

(A) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \times SR}}$
(B) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}}$
(C) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$
(D) $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$

10. In a triangle ABC with fixed base BC, the vertex A moves such that $\cos B + \cos C = 4\sin^2\frac{A}{2}$. If a, b and c denote the lengths of the sides of the triangle opposite to the angles A, B and C respectively, then

[IIT-JEE-2009 (Paper-1)]

(A) $b + c = 4a$
(B) $b + c = 2a$
(C) Locus of point A is an ellipse
(D) Locus of point A is a pair of straight lines

11. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then **[IIT-JEE-2009 (Paper-1)]**

(A) $\tan^2 x = \frac{2}{3}$
(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$
(C) $\tan^2 x = \frac{1}{3}$
(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

12. For $0<\theta<\frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6}\operatorname{cosec}\left(\theta+\frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta+\frac{m\pi}{4}\right)=4\sqrt{2}$ is(are)

[IIT-JEE-2009 (Paper-2)]

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

13. Let *ABC* be a triangle such that $\angle ACB=\frac{\pi}{6}$ and let *a*, *b* and *c* denote the lengths of the sides opposite to *A*, *B* and *C* respectively. The value(s) of *x* for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are

[IIT-JEE-2010 (Paper-1)]

(A) $-\left(2+\sqrt{3}\right)$ (B) $1+\sqrt{3}$

(C) $2+\sqrt{3}$ (D) $4\sqrt{3}$

14. Let $\theta, \varphi \in [0, 2\pi]$ be such that $2\cos\theta(1-\sin\varphi)=\sin^2\theta\left(\tan\frac{\theta}{2}+\cot\frac{\theta}{2}\right)\cos\varphi-1$, $\tan(2\pi-\theta)>0$ and $-1<\sin\theta<-\frac{\sqrt{3}}{2}$. Then φ cannot satisfy **[IIT-JEE-2012 (Paper-1)]**

(A) $0<\varphi<\frac{\pi}{2}$ (B) $\frac{\pi}{2}<\varphi<\frac{4\pi}{3}$

(C) $\frac{4\pi}{3}<\varphi<\frac{3\pi}{2}$ (D) $\frac{3\pi}{2}<\varphi<2\pi$

15. Let $f:(-1,1)\to\mathbb{R}$ be such that $f(\cos 4\theta)=\frac{2}{2-\sec^2\theta}$ for $\theta\in\left(0,\frac{\pi}{4}\right)\cup\left(\frac{\pi}{4},\frac{\pi}{2}\right)$. Then the value(s) of $f\left(\frac{1}{3}\right)$ is (are) **[IIT-JEE-2012 (Paper-2)]**

(A) $1-\sqrt{\frac{3}{2}}$ (B) $1+\sqrt{\frac{3}{2}}$

(C) $1-\sqrt{\frac{2}{3}}$ (D) $1+\sqrt{\frac{2}{3}}$

16. In a triangle *PQR*, *P* is the largest angle and cos *P* = $\frac{1}{3}$. Further the incircle of the triangle touches the sides *PQ*, *QR* and *RP* at *N*, *L* and *M* respectively, such that the lengths of *PN*, *QL* and *RM* are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are) **[JEE (Adv)-2013 (Paper-2)]**

(A) 16 (B) 18

(C) 24 (D) 22

17. In a triangle XYZ, let x, y, z be the lengths of sides opposite to the angles X, Y, Z, respectively, and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then

[JEE (Adv)-2016 (Paper-1)]

(A) Area of the triangle XYZ is $6\sqrt{6}$

(B) The radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(C) $\sin\frac{X}{2}\sin\frac{Y}{2}\sin\frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2\left(\frac{X+Y}{2}\right) = \frac{3}{5}$

18. Let α and β be non-zero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. Then which of the following is/are true? **[JEE (Adv)-2017 (Paper-2)]**

(A) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$

(C) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

19. In a triangle PQR, let $\angle PQR = 30°$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is (are) TRUE? **[JEE (Adv)-2018 (Paper-1)]**

(A) $\angle QPR = 45°$

(B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120°$

(C) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$

(D) The area of the circumcircle of the triangle PQR is 100π

20. In a non-right-angled triangle PQR, let p, q, r denote the lengths of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S, the perpendicular from P meets the side QR at E, and RS and PE intersect at O. If $p = \sqrt{3}$, $q = 1$, and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-1)]**

(A) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$

(B) Length of $RS = \frac{\sqrt{7}}{2}$

(C) Length of $OE = \frac{1}{6}$

(D) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}\left(2-\sqrt{3}\right)$

21. For non-negative integers n, let $f(n) = \dfrac{\sum_{k=0}^{n}\sin\left(\frac{k+1}{n+2}\pi\right)\sin\left(\frac{k+2}{n+2}\pi\right)}{\sum_{k=0}^{n}\sin^2\left(\frac{k+1}{n+2}\pi\right)}$ Assuming $\cos^{-1}x$ takes values in $[0, \pi]$, which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-2)]**

(A) $\lim_{n\to\infty} f(n) = \frac{1}{2}$

(B) $f(4) = \frac{\sqrt{3}}{2}$

(C) $\sin(7\cos^{-1} f(5)) = 0$

(D) If $\alpha = \tan(\cos^{-1} f(6))$, then $\alpha^2 + 2\alpha - 1 = 0$

22. Let x, y and z be positive real numbers. Suppose x, y and z are the lengths of the sides of a triangle opposite to its angles X, Y and Z, respectively. If

$$\tan\frac{X}{2}+\tan\frac{Z}{2}=\frac{2y}{x+y+z}$$

then which of the following statements is/are TRUE? **[JEE (Adv)-2020 (Paper-1)]**

(A) $2Y = X + Z$
(B) $Y = X + Z$
(C) $\tan\frac{X}{2}=\frac{x}{y+z}$
(D) $x^2+z^2-y^2=xz$

23. Consider a triangle PQR having sides of lengths p,q and r opposite to the angles P, Q and R, respectively. Then which of the following statements is (are) TRUE ? **[JEE (Adv)-2021 (Paper-2)]**

(A) $\cos P \ge 1-\frac{p^2}{2qr}$
(B) $\cos R \ge \left(\frac{q-r}{p+q}\right)\cos P+\left(\frac{p-r}{p+q}\right)\cos Q$
(C) $\frac{q+r}{p}<2\frac{\sqrt{\sin Q\sin R}}{\sin P}$
(D) If $p < q$ and $p < r$, then $\cos Q > \frac{p}{r}$ and $\cos R > \frac{p}{q}$

24. Let $PQRS$ be a quadrilateral in a plane, where $QR = 1$, $\angle PQR = \angle QRS = 70°$, $\angle PQS = 15°$ and $\angle PRS = 40°$. If $\angle RPS = \theta°$, $PQ = \alpha$ and $PS = \beta$, then the interval(s) that contain(s) the value of $4\alpha\beta\sin\theta°$ is/are

[JEE (Adv)-2022 (Paper-2)]

(A) $(0,\sqrt{2})$
(B) $(1, 2)$
(C) $(\sqrt{2}, 3)$
(D) $(2\sqrt{2}, 3\sqrt{2})$

25. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \ldots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \ldots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \ldots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE ?

[JEE (Adv)-2022 (Paper-2)]

(A) If $n = 4$, then $(\sqrt{2}-1)r < R$
(B) If $n = 5$, then $r < R$
(C) If $n = 8$, then $(\sqrt{2}-1)r < R$
(D) If $n = 12$, then $\sqrt{2}(\sqrt{3}+1)r > R$

Matrix-Match / Matrix Based Type Questions

26. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $f(x) = \sin(\pi\cos x)$ and $g(x) = \cos(2\pi\sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order : **[JEE (Adv)-2019 (Paper-2)]**

$X = \{x : f(x) = 0\}$, $Y = \{x : f'(x) = 0\}$,

$Z = \{x : g(x) = 0\}$, $W = \{x : g'(x) = 0\}$.

List-I contains the sets X, Y, Z and W. List-II contains some information regarding these sets.

List-I	**List-II**
(I) X	(P) $\supseteq\left\{\frac{\pi}{2},\frac{3\pi}{2},4\pi,7\pi\right\}$
(II) Y	(Q) An arithmetic progression

(III) Z (R) NOT an arithmetic progression

(IV) W (S) $\supseteq\left\{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}\right\}$

(T) $\supseteq\left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$

(U) $\supseteq\left\{\frac{\pi}{6}, \frac{3\pi}{4}\right\}$

Which of the following is the only CORRECT combination?

(A) (II), (R), (S) (B) (I), (Q), (U)

(C) (II), (Q), (T) (D) (I) , (P), (R)

27. Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $f(x)$ = sin(πcos$x$) and $g(x)$ = cos(2πsin$x$) be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order: **[JEE (Adv)-2019 (Paper-2)]**

$X = \{x : f(x) = 0\}$, $Y = \{x : f'(x) = 0\}$,

$Z = \{x : g(x) = 0\}$, $W = \{x : g'(x) = 0\}$.

List-I contains the sets X, Y, Z and W. List-II contains some information regarding these sets.

List-I	List-II
(I) X	(P) $\supseteq\left\{\frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi\right\}$
(II) Y	(Q) An arithmetic progression
(III) Z	(R) NOT an arithmetic progression
(IV) W	(S) $\supseteq\left\{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}\right\}$
	(T) $\supseteq\left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$
	(U) $\supseteq\left\{\frac{\pi}{6}, \frac{3\pi}{4}\right\}$

Which of the following is the only CORRECT combination?

(A) (IV), (P), (R), (S) (B) (IV), (Q), (T)

(C) (III), (R), (U) (D) (III), (P), (Q), (U)

28. Consider the following lists:

List-I	List-II
(I) $\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P) has two elements
(II) $\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$	(Q) has three elements

(III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$ (R) has four elements

(IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$ (S) has five elements

(T) has six elements

The correct option is:

(A) (I) → (P); (II) → (S); (III) → (P); (IV) → (S)

(B) (I) → (P); (II) → (P); (III) → (T); (IV) → (R)

(C) (I) → (Q); (II) → (P); (III) → (T); (IV) → (S)

(D) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)

Integer / Numerical Value Type Questions

29. Let *ABC* and *ABC′* be two non-congruent triangles with sides *AB* = 4, *AC* = *AC′* = $2\sqrt{2}$ and angle *B* = 30°. The absolute value of the difference between the areas of these triangles is **[IIT-JEE-2009 (Paper-2)]**

30. The number of all possible values of θ, where 0 < θ < π, for which the system of equations

$(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$

have a solution (x_0, y_0, z_0) with $y_0z_0 \neq 0$, is **[IIT-JEE-2010 (Paper-1)]**

31. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for *n* = 0, ± 1, ± 2 and tanθ = cot 5θ as well as sin2θ = cos4θ is **[IIT-JEE-2010 (Paper-1)]**

32. The maximum value of the expression $\frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$ is **[IIT-JEE-2010 (Paper-1)]**

33. Consider a triangle *ABC* and let *a*, *b* and *c* denote the lengths of the sides opposite to vertices *A*, *B* and *C* respectively. Suppose *a* = 6, *b* = 10 and the area of the triangle is $15\sqrt{3}$. If ∠*ACB* is obtuse and if *r* denotes the radius of the incircle of the triangle, then r^2 is equal to

[IIT-JEE-2010 (Paper-2)]

34. The positive integer value of *n* > 3 satisfying the equation $\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$ is

[IIT-JEE-2011 (Paper-1)]

35. The number of distinct solutions of the equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is **[JEE (Adv)-2015 (Paper-1)]**

36. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then, the value of $\frac{b}{a}$ is ___. **[JEE (Adv)-2018 (Paper-1)]**

37. Let $f : [0, 2] \longrightarrow \mathbb{R}$ be the function defined by

$$f(x) = (3 - \sin(2\pi x))\sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$$

If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2] : f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is _____.

[JEE (Adv)-2020 (Paper-1)]

38. In a triangle ABC, let $AB = \sqrt{23}$, and $BC = 3$ and $CA = 4$. Then the value of

$\frac{\cot A + \cot C}{\cot B}$ is ______. **[JEE (Adv)-2021 (Paper-1)]**

39. Let ABC be the triangle with $AB = 1$, $AC = 3$ and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC, then the value of r is ____________.

[JEE (Adv)-2022 (Paper-1)]

40. Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha + \beta) = \frac{1}{3}$ and $\cos(\alpha - \beta) = \frac{2}{3}$, then the greatest integer less than or equal to $\left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\cos\alpha}{\sin\beta} + \frac{\sin\beta}{\cos\alpha}\right)^2$ is _______.

[JEE (Adv)-2022 (Paper-2)]

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